

Week 3 – Linear Regression

DS3010: Introduction to Machine Learning Lab

Timing: 02:00 PM to 04:45 PM

Max Marks: 10

Instructions

1. Submit one `.ipynb` file containing all answers. The name should be `[student_name]_[rollno]_[lab].ipynb`
2. Write the questions in separate text blocks before the answers.
3. Outputs for all sub-questions should be given and the code should be executable.
4. Written interpretations must be given in *text blocks*, not as code comments.
5. Ensure that all plots include clear labels and legends for better interpretation.
6. **Use of generative AI tools (such as ChatGPT, Gemini, etc.) is strictly prohibited. Any submission found to contain AI-generated or plagiarized content will receive a score of zero, and disciplinary action.**

Getting Started

All work is to be done in a *Jupyter Notebook* on the lab machine. Everything you need is already installed — if any import below fails, inform your instructor.

1. Launch Jupyter Notebook and create a new Python 3 notebook *in the same folder* as `week-5-country-life-satisfaction.csv`, so the file can be read by name alone.
2. Run the cell below as your first cell. It checks the *Python version*, loads the libraries, and reads the dataset. Note the `index_col=0`: the first column of the CSV is an unnamed row number, not a variable.

```
import sys; print(sys.version)

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.formula.api as smf
%matplotlib inline

df = pd.read_csv("week-5-country-life-satisfaction.csv", index_col=0)
df.head()
```

Useful references — you are expected to look up the exact usage and arguments yourself:

- *Exploration*: `df.info()`, `df.describe()`, `df.corr()`, `sns.pairplot()`, `sns.heatmap()`
- *Regression*: `smf.ols("y ~ x1 + x2", data=df).fit()`, together with `.summary()`, `.params`, `.fittedvalues`, `.resid`, `.rsquared` and `.rsquared_adj`; write `C(col)` in a formula to enter a categorical predictor
- *Diagnostics*: `variance_inflation_factor`, from `statsmodels.stats.outliers_influence`
- *Evaluation*: `train_test_split`, `mean_squared_error` and `r2_score`, from `sklearn`

Background

Suppose we want to measure “quality of life” for every country. For a set of countries we have a *gold standard* measurement — average life satisfaction from the Gallup World Poll (2014–16), where respondents rate their life on a ladder from 0 to 10 — and we also have cheap, widely available *indicators* such as life expectancy and income. The idea (used by *The Economist* for its “Where-to-be-born” index) is to *train* a regression model predicting the gold standard from the indicators, and then use it to score any country for which only the indicators are available.

The dataset can be accessed here: [week-5-country-life-satisfaction.csv](#). The original problem statement and full context for this exercise is available at: [UCL POLS0013 – Supervised Scale Measurement II: Regression](#).

Use `week-5-country-life-satisfaction.csv` (134 countries) for all questions below. Its columns are:

Name, CountryCode	Country name and ISO code
Region	World Bank region (categorical)
LifeSat	Average life satisfaction, Gallup 2014–16 (<i>target measure</i>)
HDI	Human Development Index, 2018
LE	Life expectancy at birth (years)
EYS, MYS	Expected and mean years of schooling
GNI	Gross national income per capita (PPP \$)

(Suggested libraries: *pandas*, *numpy*, *matplotlib*, *seaborn*, *statsmodels*, *sklearn*.)

Questions

1. Exploring the data (2 mark)

- Load the CSV into a DataFrame. Print its *shape*, the column data types, and the number of *missing values* per column.
- Create a new column $\log\text{GNI} = \log(\text{GNI})$ using `np.log`. Plot histograms of GNI and $\log\text{GNI}$ side by side and state in one sentence why the log transform is preferred as an indicator.
- Produce a *pairwise plot* (e.g. `seaborn.pairplot`) of LifeSat, LE, EYS, MYS and $\log\text{GNI}$, and print the *correlation matrix* of these five variables (rounded to 2 decimals).
- Which indicator is most strongly correlated with LifeSat? Which *pair of indicators* is most strongly correlated with each other?

2. Fitting the model (2 marks)

- Fit a linear regression predicting LifeSat from LE, EYS, MYS and $\log\text{GNI}$. Print the full summary (coefficients, standard errors, *p*-values, R^2 and adjusted R^2).
- Write down the fitted regression equation with the estimated coefficients.
- Which indicators most strongly predict life satisfaction? Which ones are *not statistically significant*?
- One coefficient has a sign that makes *little substantive sense*. Identify it and explain in 2–3 sentences how this can happen even though that indicator was positively correlated with LifeSat in Q1.

3. Comparing with an existing index (2 marks)

- EYS and MYS are strongly correlated and largely measure the same concept. Drop the one whose coefficient made less substantive sense in Q2, *refit* the regression. Call this reduced model `model2`; use it for Questions 3 and 4.
- The dataset contains HDI, which is built from these same indicators using a fixed formula. Compute the correlation between the *fitted values* of `model2` and HDI.
- Compute the correlation between LifeSat and HDI.
- Draw two scatter plots (fitted values vs. HDI; LifeSat vs. HDI)

4. Looking at the residuals (2 marks)

- Compute the residuals of `model2` and identify the country with the *largest positive* residual and the one with the *largest negative* residual.
- Print a table comparing these two countries on LifeSat, HDI, LE, EYS, MYS, GNI and Region.
- Plot residuals against fitted values. Does the spread of the residuals look roughly constant?

5. Adding a categorical predictor

(2 marks)

- (a). Add `Region` to `model2` as a set of *dummy variables*, using either `pd.get_dummies` with `drop_first=True` or a formula with `C(Region)`. Print the summary.
- (b). State which region is the *baseline* category and explain what a region coefficient means, holding the other indicators constant.
- (c). Which regions report notably *higher* or *lower* life satisfaction than the baseline, and which of these differences are statistically significant?
- (d). Does adding `Region` improve the model? Compare R^2 *and* adjusted R^2 , and explain in one sentence why adjusted R^2 is the fairer comparison here.

Bonus (Optional)

(+1 mark)

Split the data into a *training set* (80%) and a *test set* (20%) using `train_test_split` with a fixed `random_state`. Refit `model2` on the training set, predict on the test set, and report the *RMSE* and R^2 on both.

All the best for your lab! Happy coding!